

AP® Physics 1 FRQs

Listed by Topic

Experimental design

- 2025 Q3 - - Determine mass of block using a spring scale, forces, tension, torque
- 2024 Q2 - - Masses oscillating on a spring, oscillations/simple harmonic motion, energy, momentum
- 2023 Q2 - - Cart on an incline, kinematics
- 2022 Q3 - - Block hanging from string around wheel, energy, rotational motion
- 2021 Q2 - - Stretching a plastic rod
- 2019 Q3 - - Sphere is launched with a spring plunger, projectile motion, energy
- 2017 Q2 - - Coefficient of friction between block and board, forces, friction
- 2016 Q2 - - Testing collisions for a ball, energy

Kinematics

- 2023 Q2 - - (Experimental design) Cart on an incline, kinematics
- 2022 Q1 - - Pulley system with 2 blocks and a spring, kinematics, forces, energy, work
- 2021 Q1 - - Cyclist on ramp jumps over cars, kinematics, projectile motion, energy, work
- 2019 Q1 - - Plunger pushes block and sphere across surface, kinematics, forces, energy, work, torque, angular momentum
- 2019 Q2 - - Pulley system with 2 blocks, kinematics, forces, FBDs, tension, rotational motion
- 2016 Q3 - - Cart rolls down an incline with bumps, kinematics, energy

Projectile motion

- 2022 Q4 - - Clay and ball collide with blocks, momentum, projectile motion
- 2021 Q1 - - Cyclist on ramp jumps over cars, kinematics, projectile motion, energy, work
- 2019 Q3 - - (Experimental design) Sphere is launched with a spring plunger, projectile motion, energy
- 2017 Q4 - - Blocks launched from a slide on table, projectile motion, energy
- 2015 Q4 - - Projectile motion of two spheres, projectile motion, FBDs

Free body diagrams

- 2024 Q1 - - Block sliding on a track with loops, forces, FBDs, circular motion, energy
- 2024 Q3 - - Beam attached to a wall with a string, forces, FBDs, tension, rotational motion, torque
- 2023 Q3 - - Block and spring rotating about axle, circular motion, centripetal force, FBDs
- 2022 Q2 - - Gravitational force between planet and moons, law of gravitation, FBD's

- 2019 Q2 - - Pulley system with 2 blocks, kinematics, forces, FBDs, tension, rotational motion
- 2018 Q1 - - Spacecraft in circular orbit, circular motion, forces, FBDs, law of gravitation
- 2016 Q1 - - Wheel rolls down an incline, forces, FBDs, energy, friction
- 2015 Q1 - - Pulley system with 2 hanging blocks, forces, FBDs, tension
- 2015 Q4 - - Projectile motion of two spheres, projectile motion, FBDs

Forces

- 2025 Q3 - - (Experimental design) Determine mass of block using a spring scale, forces, tension, torque
- 2025 Q4 - - Swimmer holding block in tank of water, forces, fluids
- 2024 Q1 - - Block sliding on a track with loops, forces, FBDs, circular motion, energy
- 2024 Q3 - - Beam attached to a wall with a string, forces, FBDs, tension, rotational motion, torque
- 2024 Q5 - - Collision of two blocks, momentum, energy, forces
- 2022 Q1 - - Pulley system with 2 blocks and a spring, kinematics, forces, energy, work
- 2019 Q1 - - Plunger pushes block and sphere across surface, kinematics, forces, energy, work, torque, angular momentum
- 2019 Q2 - - Pulley system with 2 blocks, kinematics, forces, FBDs, tension, rotational motion
- 2018 Q1 - - Spacecraft in circular orbit, circular motion, forces, FBDs, law of gravitation
- 2017 Q2 - - (Experimental design) Coefficient of friction between block and board, forces, friction
- 2016 Q1 - - Wheel rolls down an incline, forces, FBDs, energy, friction
- 2015 Q1 - - Pulley system with 2 hanging blocks, forces, FBDs, tension

Law of gravitation

- 2024 Q4 - - Pendulum on different planets, law of gravitation, oscillations/simple harmonic motion, work, spring force
- 2022 Q2 - - Gravitational force between planet and moons, law of gravitation, FBD's
- 2018 Q1 - - Spacecraft in circular orbit, circular motion, forces, FBDs, law of gravitation

Friction

- 2017 Q2 - - (Experimental design) Coefficient of friction between block and board, forces, friction
- 2016 Q1 - - Wheel rolls down an incline, forces, FBDs, energy, friction
- 2015 Q3 - - Block-spring system sliding on surface, energy, work, friction

Springs

- 2024 Q2 - - (Experimental design) Masses oscillating on a spring, oscillations/simple harmonic motion, energy, momentum
- 2024 Q4 - - Pendulum on different planets, law of gravitation, oscillations/simple harmonic motion, work, spring force
- 2023 Q1 - - Cart oscillating on horizontal spring, oscillations/simple harmonic motion, energy, work
- 2023 Q3 - - Block and spring rotating about axle, circular motion, centripetal force, FBDs
- 2022 Q1 - - Pulley system with 2 blocks and a spring, kinematics, forces, energy, work
- 2022 Q5 - - Mass oscillating on a vertical spring, oscillations/simple harmonic motion, energy, work
- 2019 Q3 - - (Experimental design) Sphere is launched with a spring plunger, projectile motion, energy
- 2015 Q3 - - Block-spring system sliding on surface, energy, work, friction

Tension

- 2025 Q3 - - (Experimental design) Determine mass of block using a spring scale, forces, tension, torque
- 2024 Q3 - - Beam attached to a wall with a string, forces, FBDs, tension, rotational motion, torque
- 2022 Q1 - - Pulley system with 2 blocks and a spring, kinematics, forces, energy, work
- 2022 Q3 - - (Experimental design) Block hanging from string around wheel, energy, rotational motion
- 2021 Q5 - - 2 blocks hanging from strings on pulleys, rotational motion, torque
- 2019 Q2 - - Pulley system with 2 blocks, kinematics, forces, FBDs, tension, rotational motion
- 2015 Q1 - - Pulley system with 2 hanging blocks, forces, FBDs, tension

Circular motion

- 2024 Q1 - - Block sliding on a track with loops, forces, FBDs, circular motion, energy
- 2023 Q3 - - Block and spring rotating about axle, circular motion, centripetal force, FBDs
- 2018 Q1 - - Spacecraft in circular orbit, circular motion, forces, FBDs, law of gravitation

Energy

- 2025 Q1 - - Block is dropped on a moving cart, energy, momentum
- 2025 Q2 - - Block and spring on an incline, energy
- 2024 Q1 - - Block sliding on a track with loops, forces, FBDs, circular motion, energy

- 2024 Q2 - - (Experimental design) Masses oscillating on a spring, oscillations/simple harmonic motion, energy, momentum
- 2024 Q5 - - Collision of two blocks, momentum, energy, forces
- 2023 Q1 - - Cart oscillating on horizontal spring, oscillations/simple harmonic motion, energy, work
- 2023 Q4 - - Block hanging from string around pulley, rotational motion, torque, angular momentum, energy, work
- 2023 Q5 - - Sphere and rod rotating about axle, rotational motion, torque, energy, work
- 2022 Q1 - - Pulley system with 2 blocks and a spring, kinematics, forces, energy, work
- 2022 Q3 - - (Experimental design) Block hanging from string around wheel, energy, rotational motion
- 2022 Q5 - - Mass oscillating on a vertical spring, oscillations/simple harmonic motion, energy, work
- 2021 Q1 - - Cyclist on ramp jumps over cars, kinematics, projectile motion, energy, work
- 2021 Q4 - - Cylinder rolls and block slides down an incline, energy, work, rotational motion
- 2019 Q1 - - Plunger pushes block and sphere across surface, kinematics, forces, energy, work, torque, angular momentum
- 2019 Q3 - - (Experimental design) Sphere is launched with a spring plunger, projectile motion, energy
- 2018 Q5 - - Block dropped on another oscillating block, energy, momentum, oscillations/simple harmonic motion
- 2017 Q4 - - Blocks launched from slides on tables, projectile motion, energy
- 2016 Q1 - - Wheel rolls down an incline, forces, FBDs, energy, friction
- 2016 Q2 - - (Experimental design) Testing collisions for a ball, energy
- 2016 Q3 - - Cart rolls down an incline with bumps, kinematics, energy
- 2015 Q3 - - Block-spring system sliding on surface, energy, work, friction

Work

- 2024 Q4 - - Pendulum on different planets, law of gravitation, oscillations/simple harmonic motion, work, spring force
- 2023 Q1 - - Cart oscillating on horizontal spring, oscillations/simple harmonic motion, energy, work
- 2023 Q4 - - Block hanging from string around pulley, rotational motion, torque, angular momentum, energy, work
- 2023 Q5 - - Sphere and rod rotating about axle, rotational motion, torque, energy, work
- 2022 Q1 - - Pulley system with 2 blocks and a spring, kinematics, forces, energy, work
- 2022 Q5 - - Mass oscillating on a vertical spring, oscillations/simple harmonic motion, energy, work
- 2021 Q1 - - Cyclist on ramp jumps over cars, kinematics, projectile motion, energy, work
- 2021 Q4 - - Cylinder rolls and block slides down an incline, energy, work, rotational motion

- 2019 Q1 - - Plunger pushes block and sphere across surface, kinematics, forces, energy, work, torque, angular momentum
- 2015 Q3 - - Block-spring system sliding on surface, energy, work, friction

Momentum

- 2025 Q1 - - Block is dropped on a moving cart, energy, momentum
- 2024 Q2 - - (Experimental design) Masses oscillating on a spring, oscillations/simple harmonic motion, energy, momentum
- 2024 Q5 - - Collision of two blocks, momentum, energy, forces
- 2022 Q4 - - Clay and ball collide with blocks, momentum, projectile motion
- 2021 Q3 - - Student pushes disk which hits a block, momentum
- 2018 Q5 - - Block dropped on another oscillating block, energy, momentum, oscillations/simple harmonic motion

Rotational motion

- 2024 Q3 - - Beam attached to a wall with a string, forces, FBDs, tension, rotational motion, torque
- 2023 Q4 - - Block hanging from string around pulley, rotational motion, torque, angular momentum, energy, work
- 2023 Q5 - - Sphere and rod rotating about axle, rotational motion, torque, energy, work
- 2022 Q3 - - (Experimental design) Block hanging from string around wheel, energy, rotational motion
- 2021 Q4 - - Cylinder rolls and block slides down an incline, energy, work, rotational motion
- 2021 Q5 - - 2 blocks hanging from strings on pulleys, rotational motion, torque
- 2019 Q2 - - Pulley system with 2 blocks, kinematics, forces, FBDs, tension, rotational motion
- 2018 Q3 - - Disk spinning on axle, rotational motion, torque
- 2017 Q3 - - Disk collides with pivoting rod, rotational motion, torque, angular momentum

Torque

- 2025 Q3 - - (Experimental design) Determine mass of block using a spring scale, forces, tension, torque
- 2024 Q3 - - Beam attached to a wall with a string, forces, FBDs, tension, rotational motion, torque
- 2023 Q4 - - Block hanging from string around pulley, rotational motion, torque, angular momentum, energy, work
- 2023 Q5 - - Sphere and rod rotating about axle, rotational motion, torque, energy, work
- 2021 Q5 - - 2 blocks hanging from strings on pulleys, rotational motion, torque
- 2019 Q1 - - Plunger pushes block and sphere across surface, kinematics, forces, energy, work, torque, angular momentum

- 2018 Q3 - - Disk spinning on axle, rotational motion, torque
- 2017 Q3 - - Disk collides with pivoting rod, rotational motion, torque, angular momentum

Angular momentum

- 2023 Q4 - - Block hanging from string around pulley, rotational motion, torque, angular momentum, energy, work
- 2019 Q1 - - Plunger pushes block and sphere across surface, kinematics, forces, energy, work, torque, angular momentum
- 2017 Q3 - - Disk collides with pivoting rod, rotational motion, torque, angular momentum

Oscillations / simple harmonic motion

- 2024 Q2 - - (Experimental design) Masses oscillating on a spring, oscillations/simple harmonic motion, energy, momentum
- 2024 Q4 - - Pendulum on different planets, law of gravitation, oscillations/simple harmonic motion, work, spring force
- 2023 Q1 - - Cart oscillating on horizontal spring, oscillations/simple harmonic motion, energy, work
- 2022 Q5 - - Mass oscillating on a vertical spring, oscillations/simple harmonic motion, energy, work
- 2018 Q5 - - Block dropped on another oscillating block, energy, momentum, oscillations/simple harmonic motion

Fluids

The fluids questions from AP Physics 2 before 2025 are also included. Only parts of those questions may be relevant to AP Physics 1 as noted below.

- 2025 Q4 - - Swimmer holding block in tank of water, forces, fluids
- 2023 Q3 - - Tank of water with pipe and two blocks, fluids, forces, buoyant force, flow, Bernoulli's equation, continuity equation
- 2022 Q1 - - (Part a) Block in tank of water hanging from string, fluids, forces, FBDs, buoyant force
- 2021 Q2 - - (Parts b and c) Balloon held underwater, fluids, density, pressure, forces, buoyant force
- 2019 Q4 - - (parts b and c) Air bubble rises in water, fluids, pressure, forces, buoyant force, energy, work
- 2018 Q4 - - (Parts a and c) Boat carrying steel bars, fluids, forces, buoyant force, flow, continuity equation
- 2017 Q1 - - Water flowing through pipe, fluids, flow, continuity equation, Bernoulli's equation, conservation of energy, forces, FBDs, buoyant force